

ENTITY RELATIONSHIP (ER) DIAGRAM AND RELATIONAL SCHEMA

Hospital Management System

1. Objective

The selected application is a Hospital Management System. The database stores patient information, appointments, doctors, departments, admissions, medical records, bills, prescriptions, and medicines. The design also represents the many-to-many relationship between prescriptions and medicines through an associative relation.

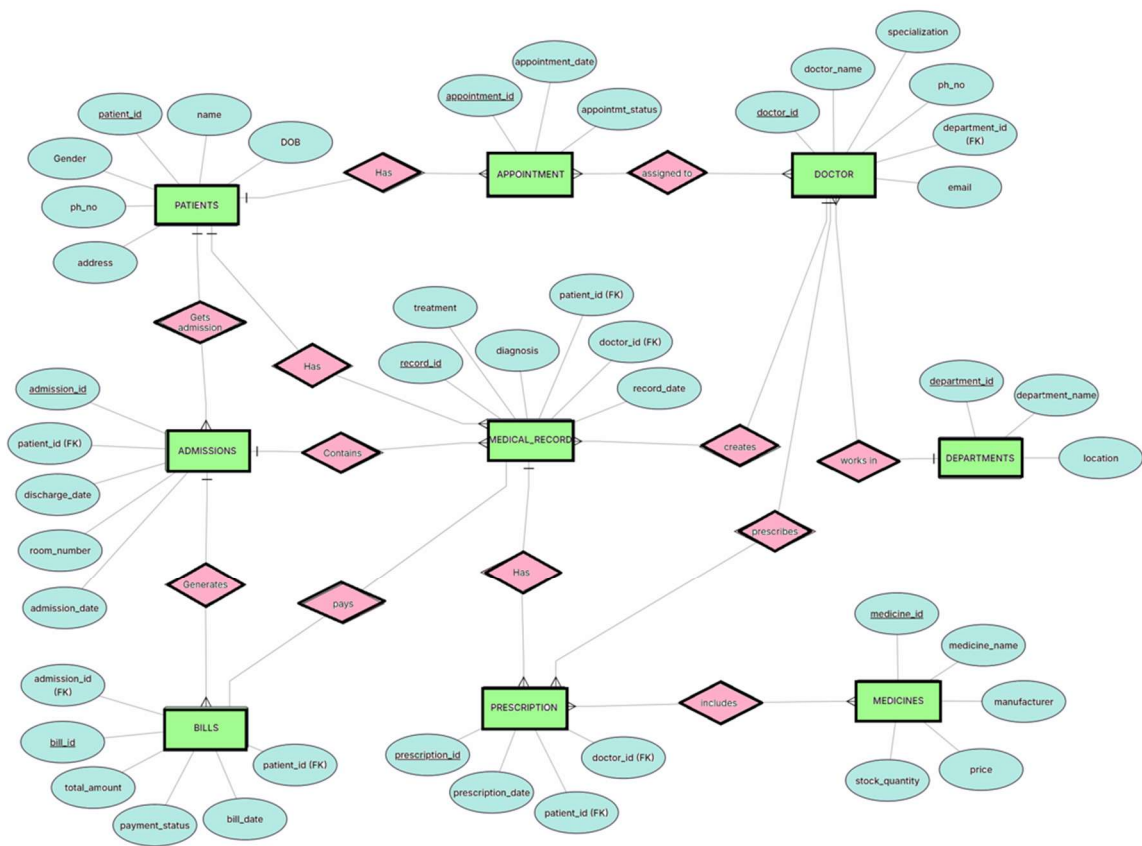
2. Entities and Attributes

The following entities are used in the ER model. PK denotes Primary Key and FK denotes Foreign Key.

Entity	Primary Key	Other Attributes / Foreign Keys
PATIENTS	patient_id (PK)	name; DOB; Gender; ph_no; address
APPOINTMENT	appointment_id (PK)	appointment_date; appointment_status; patient_id (FK); doctor_id (FK)
DOCTOR	doctor_id (PK)	doctor_name; specialization; ph_no; email; department_id (FK)
DEPARTMENTS	department_id (PK)	department_name; location
ADMISSIONS	admission_id (PK)	admission_date; discharge_date; room_number; patient_id (FK); doctor_id (FK)
MEDICAL_RECORD	record_id (PK)	diagnosis; treatment; record_date; patient_id (FK); doctor_id (FK)
BILL	bill_id (PK)	total_amount; payment_status; bill_date; patient_id (FK); admission_id (FK)
PRESCRIPTION	prescription_id (PK)	prescription_date; patient_id (FK); doctor_id (FK)
MEDICINES	medicine_id (PK)	medicine_name; manufacturer; price; stock_quantity
PRESCRIPTION_MEDICINE	PK = (prescription_id, medicine_id)	prescription_id (FK); medicine_id (FK); quantity

3. ER Diagram

The ER diagram below represents the entities, attributes, relationships, and the overall structure of the Hospital Management System.



4. Relationships and Cardinalities

Entities	Relationship	Cardinality	Explanation
PATIENTS — APPOINTMENT	Has	1 : M	One patient can have many appointments; each appointment belongs to one patient.
DOCTOR — APPOINTMENT	Assigned to	1 : M	One doctor can be assigned to many appointments; each appointment is assigned to one doctor.
PATIENTS — ADMISSIONS	Gets admission	1 : M	One patient can have multiple admissions over time; each admission belongs to one patient.
DOCTOR — ADMISSIONS	Handles	1 : M	One doctor can handle multiple admissions; each admission records one responsible doctor.
DEPARTMENTS — DOCTOR	Works in	1 : M	One department can have multiple doctors; each doctor belongs to one department.
PATIENTS — MEDICAL_RECORD	Has	1 : M	A patient can have multiple medical records; each medical record belongs to one patient.
DOCTOR — MEDICAL_RECORD	Creates	1 : M	A doctor can create many medical records; each record is associated with one doctor.
ADMISSIONS — MEDICAL_RECORD	Contains	1 : M	An admission may contain multiple medical records. Records are associated with the relevant patient and doctor.
PATIENTS — PRESCRIPTION	Has	1 : M	A patient can receive multiple prescriptions; each prescription belongs to one patient.
DOCTOR — PRESCRIPTION	Prescribes	1 : M	A doctor can issue multiple prescriptions; each prescription is issued by one doctor.
MEDICAL_RECORD — PRESCRIPTION	Has	1 : M	A medical record can be associated with multiple prescriptions where required by the clinical workflow.
ADMISSIONS — BILL	Generates	1 : M	An admission can generate one or more billing records; each bill references the related admission.
PATIENTS — BILL	Pays	1 : M	A patient can have multiple bills; each bill references one patient.

PRESCRIPTION — MEDICINES	Includes	M : N	A prescription may contain many medicines and a medicine may appear on many prescriptions; resolved by PRESCRIPTION_MEDICINE.
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5. Relational Schema

The ER model is converted into relations using the standard mapping rules: entities become tables, identifying attributes become primary keys, foreign keys are placed on the many-side of 1:M relationships, and the M:N relationship is converted into an associative table.

Relation	Relational Schema
PATIENTS	PATIENTS(patient_id PK, name, DOB, Gender, ph_no, address)
DEPARTMENTS	DEPARTMENTS(department_id PK, department_name, location)
DOCTOR	DOCTOR(doctor_id PK, doctor_name, specialization, ph_no, email, department_id FK → DEPARTMENTS.department_id)
APPOINTMENT	APPOINTMENT(appointment_id PK, appointment_date, appointment_status, patient_id FK → PATIENTS.patient_id, doctor_id FK → DOCTOR.doctor_id)
ADMISSIONS	ADMISSIONS(admission_id PK, admission_date, discharge_date, room_number, patient_id FK → PATIENTS.patient_id, doctor_id FK → DOCTOR.doctor_id)
MEDICAL_RECORD	MEDICAL_RECORD(record_id PK, diagnosis, treatment, record_date, patient_id FK → PATIENTS.patient_id, doctor_id FK → DOCTOR.doctor_id)
BILL	BILL(bill_id PK, total_amount, payment_status, bill_date, patient_id FK → PATIENTS.patient_id, admission_id FK → ADMISSIONS.admission_id)
PRESCRIPTION	PRESCRIPTION(prescription_id PK, prescription_date, patient_id FK → PATIENTS.patient_id, doctor_id FK → DOCTOR.doctor_id)
MEDICINES	MEDICINES(medicine_id PK, medicine_name, manufacturer, price, stock_quantity)
PRESCRIPTION_MEDICINE	PRESCRIPTION_MEDICINE(prescription_id PK/FK → PRESCRIPTION.prescription_id, medicine_id PK/FK → MEDICINES.medicine_id, quantity)

6. Primary Key and Foreign Key Identification

Table	Primary Key (PK)	Foreign Key(s) (FK)
PATIENTS	patient_id	—
DEPARTMENTS	department_id	—
DOCTOR	doctor_id	department_id → DEPARTMENTS(department_id)
APPOINTMENT	appointment_id	patient_id → PATIENTS(patient_id); doctor_id → DOCTOR(doctor_id)
ADMISSIONS	admission_id	patient_id → PATIENTS(patient_id); doctor_id → DOCTOR(doctor_id)
MEDICAL_RECORD	record_id	patient_id → PATIENTS(patient_id); doctor_id → DOCTOR(doctor_id)
BILL	bill_id	patient_id → PATIENTS(patient_id); admission_id → ADMISSIONS(admission_id)
PRESCRIPTION	prescription_id	patient_id → PATIENTS(patient_id); doctor_id → DOCTOR(doctor_id)
MEDICINES	medicine_id	—
PRESCRIPTION_MEDICINE	(prescription_id, medicine_id)	prescription_id → PRESCRIPTION(prescription_id); medicine_id → MEDICINES(medicine_id)

7. Brief Relationship Explanation

- Patient–Appointment: The relationship is 1:M because a patient may schedule several appointments, while each appointment is associated with one patient.

- Doctor–Appointment: The relationship is 1:M because a doctor may handle many appointments, while each appointment is assigned to one doctor.
- Patient–Admission: The relationship is 1:M because a patient may be admitted more than once over time, while an admission belongs to one patient.
- Department–Doctor: The relationship is 1:M because a department can contain multiple doctors, while a doctor is assigned to one department in this design.
- Patient/Doctor–Medical Record: Both are 1:M relationships. A patient can accumulate multiple medical records and a doctor can create multiple records.
- Patient/Doctor–Prescription: Both are 1:M relationships. A patient may receive multiple prescriptions and a doctor may issue multiple prescriptions.
- Admission–Bill and Patient–Bill: Both are modeled as 1:M, allowing multiple billing records to be associated with an admission or patient.
- Prescription–Medicine: This is M:N. One prescription can contain multiple medicines and one medicine can be included in many prescriptions. The PRESCRIPTION_MEDICINE table resolves this M:N relationship and stores quantity.

8. ERD Reference

The following image is the ERD supplied with the assignment and is included as a reference for the design above.

